HTML FUNDAMENTALS

Lecture Notes — English Edition

Web Page Structure • Tags • Attributes • Images • Links • Lists • Forms • Embedded Content

Prepared from the uploaded lecture transcript by Abhishek Purohit (Oct 21). The notes preserve the lecture's concepts, examples, and emphasis while organizing them for revision.

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Source note: this is a cleaned study guide, not a word-for-word transcript. Where the transcript gives only a brief mention or the available text is unclear, the notes avoid inventing detailed lecture content.

1. What HTML Is

HTML stands for HyperText Markup Language. The lecturer describes HTML as the skeleton or structure of a web page. More simply, HTML is used to display and organize content on a web page.

For example, the visible parts of a website—such as a logo, search bar, buttons, text, forms, and tables—are represented using HTML. CSS is used to style the page and make it look more attractive.

Core idea

HTML describes the content and structure to render. It is not used to write programming logic in the way C, C++, or Java are.

- HTML does not use variables, loops, if-else statements, or switch statements for page structure.
- HTML provides predefined tags for common elements such as paragraphs, headings, buttons, inputs, forms, and tables.
- The developer decides which elements the UI needs, then uses the corresponding tags.

2. Creating an HTML File

Create a file with the .html extension, such as `demo.html`. The filename itself is your choice; the extension identifies it as an HTML document.

You can write HTML in editors such as VS Code, Sublime Text, Brackets, Atom, or even Notepad. The lecturer recommends using a code editor such as VS Code rather than Notepad.

First steps

- Create a project folder.
- Create a file such as demo.html.
- Open the file in a code editor.
- Write HTML tags inside it.
- Open the HTML file in a browser to see the rendered result.

3. Tags and Basic Rules

A tag is written using angle brackets. The tag name goes between the brackets. The lecturer emphasizes that HTML has predefined tags for standard elements.

```
<p>This is a paragraph.</p>
<button>Click me</button>
```

In the example, and mark the paragraph's opening and closing tags. The text between them is the content.

Case sensitivity

The lecturer says HTML tag names are case-insensitive: uppercase and lowercase forms are generally interpreted the same. However, consistent lowercase tag names are recommended for readability and good practice.

HTML and errors

The lecturer describes HTML as 'error-free' in the sense that browsers often still render a page even when markup is malformed. This can be misleading during development: incorrect markup may cause content to appear in the wrong place or produce UI bugs instead of a clear compile-time error. Check syntax carefully.

4. Container Tags vs. Empty Tags

Type	Description	Examples mentioned
Container tag	Has an opening tag and a corresponding closing tag. The content is placed between them.	`<form>...</form>`, `<body>...</body>`
Empty tag	Does not wrap content in a separate opening/closing pair; it is used on its own.	` `, `<img>`

A closing tag repeats the tag name with a slash after the opening angle bracket. Attributes belong in the opening tag; they are not repeated in the closing tag.

5. Basic HTML Document Structure

The lecturer introduces the <html> tag as the container for the whole document, divided into the <head> and <body> sections.

```
<!DOCTYPE html>
<html>
  <head>
    <title>My First Page</title>
  </head>
  <body>
    <p>Hello World</p>
  </body>
</html>
```

Element	Purpose
<html>	Wraps the HTML document.
<head>	Contains page information such as the document title; the title appears in the browser tab.
<title>	Sets the page title shown in the browser tab.
<body>	Contains the main content displayed in the page area.
<p>	Displays paragraph text.

The lecturer also mentions a favicon as the small icon shown with a page in the browser tab; how to add one is left for a later session.

6. Browser Rendering, DOM and Inspect

HTML does not need a traditional compiler like C or Java. The browser reads the HTML and renders the page. The lecturer mentions the DOM (Document Object Model) as part of the browser's behind-the-scenes process; it is to be discussed in a later session.

Using the browser's Inspect / DevTools Elements panel, a developer can view the page's HTML and temporarily edit text or styles to test how a UI might look.

Important limitation

An edit made in DevTools changes the page only in that browser session on that device. It does not update the original website or its server-side data. Reloading the page normally restores the original server-provided content. The lecturer demonstrates this using a product price as an example.

This feature is useful for debugging and temporary visual testing; it is not a way to change the real price of a product.

7. Paragraph vs. Span

Tag	Behavior described in the lecture	Use
<p>	Starts on a new line and occupies the full available line width (block-like behavior).	For a paragraph or block of text.
	Does not take the full line width; multiple spans can appear on the same line (inline behavior).	For a short inline piece of text.

```
<p>Hello World</p>
<p>Hi</p>
<span>Hello</span> <span>World</span>
```

Neither is universally better. Choose according to the layout requirement.

8. Heading Tags

HTML provides six heading tags: <h1> through <h6>. The lecturer explains that <h1> is the largest and the size gradually decreases through <h6>.

```
<h1>Heading 1</h1>
<h2>Heading 2</h2>
<h3>Heading 3</h3>
<h4>Heading 4</h4>
<h5>Heading 5</h5>
```

```
<h6>Heading 6</h6>
```

There are no predefined tags such as <h1.5> or <h2.5>. If a design needs a custom font size, the lecturer says to use CSS.

9. Attributes and Alignment

A tag defines the element; an attribute supplies additional information or configuration for that element. Attributes are written inside the opening tag as a name and value.

```
<p align="center">Hello World</p>
<img src="photo.png" width="500" height="300">
```

In the first example, `p` is the tag name and `align` is the attribute. In the second, `img` is the tag and `src`, `width`, and `height` are attributes.

The lecture demonstrates left, center, and right alignment using the legacy `align` attribute. The lecturer explicitly recommends CSS for alignment in real development because alignment is a styling concern.

Attribute reminders

- Attributes are written in the opening tag.
- An attribute is not a tag.
- The closing tag only needs the tag name, not the opening tag's attributes.
- A tag may have multiple attributes.

10. Images and Relative Paths

Use the <img> empty tag to display an image. Its `src` attribute specifies the image path.

```
<img src="photo.png" width="500" height="300">
```

If the HTML file and image are in the same folder, the filename can be used directly. If the image is in an `images` subfolder, use a relative path such as `images/photo.png`.

Example folder layout

```
project/
  demo.html
  images/
    photo.png
```

From demo.html, the relative image path is `images/photo.png`.

When the HTML file is inside another folder

```
project/
  images/
    photo.png
  webpage/
    demo.html
```

From `webpage/demo.html`, use `../images/photo.png`. The path segment `../` moves up one folder level. Use it repeatedly to move up multiple levels.

Width, height and aspect ratio

- If width and height are omitted, the image is normally displayed at its intrinsic dimensions.
- The lecture uses numeric width and height values as pixels, e.g. width='500' and height='300'.
- Very large dimensions can create scrollbars; enlarging a small image can make it pixelated.
- The required aspect ratio depends on the design. The lecture notes that both dimensions do not always have to be specified.

Path recommendation from the lecture: use relative paths inside the application folder rather than machine-specific absolute file paths. A deployed server will not share your local computer's C: or D: drive structure.

The lecturer also recommends keeping application files—web pages and images—together inside a single project folder so the whole application can be deployed as a unit.

11. Anchor Links and href

Use the <a> (anchor) tag to display a link. The visible text is placed between the opening and closing anchor tags. The `href` attribute specifies the destination.

```
<a href="https://www.flipkart.com">Visit Flipkart</a>
```

In this example:

- a is the tag name.
- `href` is an attribute, not a tag.
- Visit Flipkart is the visible link text.
- The URL in href is the destination opened when the user clicks the link.

The lecturer notes that href can point to an external website or to another HTML file in the same project, such as `home.html` or `contact-us.html`.

12. Lists and Tables

The later part of the transcript discusses lists, including unordered lists (`<ul>`) and definition/description lists (`<dl>`). The lecturer notes that some definition-list content can also be represented using a `<ul>` when the simpler structure meets the requirement.

The transcript also mentions tables as one of the predefined HTML elements used to display data in rows and columns. The available transcript excerpt does not provide a complete table-tag demonstration, so detailed table syntax is not added here.

13. Forms and Input Controls

Forms and input controls are introduced as examples of elements that have predefined HTML tags. The lecture mentions forms, buttons, search fields, password fields, dropdowns, and radio buttons.

The transcript emphasizes that a container tag such as `<form>` has an opening and closing tag. It does not provide a complete, reliably recoverable form example in the available text, so this section stays at the level supported by the source.

14. iframe and Embedded Content

Near the end of the session, the lecturer demonstrates embedding a Google Map. Google Maps provides an `<iframe>` snippet through its Share / Embed a map option.

The lecturer advises copying the supplied iframe code without changing its source URL or other generated details. Width and height can be adjusted to control the displayed dimensions.

```
<iframe src="..." width="600" height="450"></iframe>
```

The purpose of an iframe is to show content from another page or service inside your own page. The example source URL is omitted above because it is generated for a specific map.

15. Quick Revision and MCQs

Question	Quick answer
What does HTML stand for?	HyperText Markup Language.
Main purpose of HTML?	Create the structure/skeleton and display content on a web page.
What is a tag?	A predefined markup element written using angle brackets.
Container vs. empty tag?	Container has opening and closing tags; empty tag is used without a content pair.
What goes in head?	Page information such as the title.
What goes in body?	Main visible page content.
Does HTML need a traditional compiler?	No; the browser reads and renders the HTML.
What does DOM stand for?	Document Object Model.
p vs. span?	p starts on a new line/full-width block; span is inline.
How many heading tags?	Six: h1 through h6.
What is an attribute?	Extra information/configuration written in the opening tag.
Which attribute gives an image path?	src.
Which attribute sets a link destination?	href.
What does ../ do in a relative path?	Moves up one folder level.
Which tag creates a link?	a (anchor).
Which tag embeds external page content?	iframe.

Practice MCQs

1. What is the full form of HTML?

- A. HighText Machine Language
- B. HyperText Markup Language
- C. Hyper Tool Multi Language
- D. Home Tool Markup Language

Answer: B — HyperText Markup Language.

2. Which tag is used for a paragraph?

- A. `<para>`
- B. `<p>`
- C. `<text>`
- D. `<pg>`

Answer: B — <p>.

3. Which element sets the browser-tab title?

- A. body
- B. title
- C. h1
- D. span

Answer: B — title.

4. Which tag is an empty tag in the lecture examples?

- A. form
- B. body
- C. img
- D. html

Answer: C — img.

5. Which tag is inline according to the lecture?

- A. p
- B. span
- C. body
- D. form

Answer: B — span.

6. Which tag is the largest heading by default?

- A. h6
- B. h3
- C. h2
- D. h1

Answer: D — h1.

7. Which attribute specifies the image source?

- A. href
- B. src
- C. align
- D. target

Answer: B — src.

8. Which path moves up one directory level?

- A. ./
- B. ../
- C. //
- D. /..

Answer: B — ../.

9. In an anchor tag, what does href specify?

- A. Visible text
- B. Link destination
- C. Font size
- D. Image width

Answer: B — the destination URL or path.

10. What happens when you edit a product price in DevTools Elements?

- A. The server price changes permanently
- B. The price changes for all users
- C. The edit is temporary on the local page
- D. The database is automatically updated

Answer: C — the temporary browser-side rendering changes; reloading restores the original.

11. What does DOM stand for?

- A. Data Object Mode
- B. Document Object Model
- C. Display Order Markup
- D. Document Output Method

Answer: B — Document Object Model.

12. Which tag was used in the lecture to embed a Google Map?

- A. map
- B. object
- C. iframe
- D. link

Answer: C — iframe.

16. Final Checklist

- Know the purpose of HTML and why it is called the page's structure/skeleton.

- Create and open a .html file in a code editor and browser.
- Recognize tags, opening/closing tags, and attributes.
- Know the document structure: html, head, title, and body.
- Explain browser rendering, DOM, and the temporary nature of DevTools edits.
- Distinguish p, span, and h1–h6.
- Use src, width, height, href, and relative paths appropriately.
- Explain why ../ is used when the image is in a sibling folder.
- Recognize ul, dl, form, input controls, and iframe from the lecture.
- Practice the MCQs and review any tag or path syntax without looking.